Software Requirements Specification

for

UG/PG PROJECT ALLOCATION

Version 1.0 approved

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Group 5

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Revision History

Name	Date	Reason For Changes	Version
SeLabGroup5	11-01-2026		1.0.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for the UG/PG Project Allocation System. This system supports academic institutions in allocating projects, where faculty provide project supervision to students at the undergraduate and postgraduate levels.

The system allows students to request supervision, and faculty and coordinators to review and manage these requests within defined allocation timelines. This document serves as a reference for all stakeholders involved in the design, development, and evaluation of the software.

1.2 Document Conventions

This SRS document follows standard IEEE-style software requirements structuring. Functional requirements are labeled using the prefix FR for clarity. Numbering conventions are hierarchical (e.g., 1, 1.1) to indicate section relationships.

All mandatory requirements are stated using the keyword "shall", while optional or recommended requirements use the terms "should" or "may". Priority levels are explicitly stated for individual requirements where applicable and are not inherited implicitly. No special fonts, colors, or typographical emphasis are used beyond standard bold headings for readability.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for stakeholders involved in the development, deployment, and evaluation of the UG/PG Project Allocation System. The primary audience includes:

- ***Developers and Designers**, who will use the requirements to design and implement system features.*

- ***Testers**, who will validate that the system meets the specified requirements.*

- ***Project Managers and Academic Coordinators**, who will verify that the system aligns with institutional objectives and allocation policies.*

- ***Institutional Administrators**, who will oversee deployment, configuration, and operational use of the system.*

The document is organized to present an overview of the system. Readers unfamiliar with the system should begin with the introductory sections to understand the context and purpose before referring to the requirement sections for implementation or validation activities.

1.4 Product Scope

The UG/PG Project Allocation System is a web-based application designed to manage the assignment of projects to students in a fair and transparent manner. It replaces traditional manual allocation processes that often lead to inefficiencies, biases, and uneven faculty workload distribution.

The system enables students to request supervision, faculty to review and respond to requests, and coordinators to monitor the allocation process within defined academic guidelines. By streamlining these interactions, the system reduces administrative effort, promotes balanced supervision capacity, and ensures better alignment between student interests and faculty expertise.

This solution supports institutional goals of improving academic quality, student satisfaction, and operational efficiency, while providing a scalable and reusable allocation framework for future academic cycles.

1.5 References

The following documents and technologies are referenced in the design and implementation of this system. These references provide background information on frameworks, platforms, and tools used in the system.

- ***React***
  https://react.dev/
- ***Spring Framework***
  https://spring.io/projects/spring-framework
- ***Spring Boot***
  https://spring.io/projects/spring-boot
- ***Spring Security***
  https://spring.io/projects/spring-security
- ***MariaDB***
  https://mariadb.org/
- ***Redis***
  https://redis.io/
- ***shadcn***
  https://ui.shadcn.com/

2. Overall Description

2.1 Product Perspective

*The **UG/PG Project Allocation System** is a web-based application developed to automate and streamline the process of allocating academic projects to undergraduate and postgraduate student teams under faculty supervision. The system follows a **three-way handshake allocation model** to ensure fairness, flexibility, and mutual consent between students and faculty.*

*This system is a **new, standalone product** and does not replace any existing software system directly. It is intended to be used within an academic institution and integrates students, faculty, and administrators onto a single platform. The application will be developed using **React for the frontend** and **Java (Spring Boot) for the backend**, with a centralized database for persistent storage.*

The product acts as an intermediary between:

- *Student teams seeking project guidance*
- *Faculty offering project supervision*
- *Department administration overseeing the process*

2.2 Product Functions

At a high level, the system provides the following major functions:

- *User authentication and role-based access control*
- *Student profile creation and management (UG/PG)*
- *Faculty profile management with domain and expertise details*
- *Formation of student teams*
- *Designation of one Team Leader per team*
- *Project creation and publishing by faculty*
- *Team Leader applying to interested projects on behalf of the team*
- *Faculty viewing details of teams who applied for projects under them*
- *Faculty accepting selected teams and sending allotment offers*
- *Team Leader choosing to accept one project or wait for other offers*
- *Finalization of project allotment after team confirmation*
- *Automatic slot and conflict management*
- *Notifications for application, acceptance, and confirmation events*
- *Administrative monitoring, override, and reporting*

*The system enforces a strict **three-stage workflow**:*
Team Application → Faculty Acceptance → Team Confirmation

2.3 User Classes and Characteristics

1. Students (UG / PG)

- *Undergraduate and Postgraduate students.*
- *Can create and update personal profiles.*

- *Can form teams*
- *Each team has exactly one Team Leader.*
- *Only the Team Leader can apply to faculty and confirm allotment.*
- *Basic to moderate technical proficiency.*
- *Access restricted to their own team and applications.*

2. Faculty / Professor

- *Academic staff who supervise student projects.*
- *Can manage profiles, define expertise domains, and publish projects.*
- *Receive and review multiple team applications.*
- *Can accept selected teams and send allotment offers.*
- *Moderate technical proficiency.*

3. Department Coordinator / Admin

- *Oversees the entire project allocation process.*
- *Manages timelines, system configurations, and supervision limits.*
- *Resolves conflicts and can override allocations when necessary.*
- *High system privileges.*

4. System Administrator

- *Manages system infrastructure and technical operations.*
- *Maintains user accounts, roles, and security settings.*
- *Responsible for backups, logs, and system availability.*
- *Very high technical proficiency.*

2.4 Operating Environment

The system will operate in the following environment:

- ***Frontend:*** *React (Web Browser)*
- ***Backend:*** *Java with Spring Boot*
- ***Client Devices:*** *Desktop, Laptop, Tablet*
- ***Browsers:*** *Chrome, Firefox, Edge, Safari*
- ***Server OS:*** *Linux / Windows*
- ***Database:*** *MariaDB*
- ***Network:*** *Internet / Institutional LAN / Wifi*

The system is expected to run on standard web servers and coexist with existing institutional IT infrastructure.

2.5 Design and Implementation Constraints

The system design and implementation are constrained by:

- *Institutional authentication and security policies*
- *Role-based access control requirements*
- *Only Team Leader is permitted to apply and confirm*
- *Three-way handshake workflow must be strictly followed*
- *A team can be finally allotted to only one project*
- *Faculty supervision is limited by available slots*
- *A student cannot be part of more than one team*
- *Technologies must be React (frontend) and Java Spring Boot (backend)*
- *Data privacy regulations regarding student and faculty information*

The system must prevent:

- *Duplicate project allotments*
- *Slot overbooking*
- *Unauthorized access and actions*
- *Inconsistent team or allocation states*

2.6 User Documentation

The following user documentation will be provided:

- *Student User Manual*
- *Faculty User Manual*
- *Department Coordinator/Admin Guide*
- *System Administrator Guide*
- *Online Help and Tooltips*

2.7 Assumptions and Dependencies

Assumptions

- *All users have valid institutional credentials.*
- *Students will form teams before applying to faculty.*
- *Each team will have exactly one Team Leader.*
- *Faculty will regularly review and respond to applications.*
- *Users have basic knowledge of web-based systems.*
- *Stable internet connectivity is available.*

Dependencies

- *Availability of faculty and project data*
- *Proper functioning of authentication services*
- *Database server availability and reliability*
- *Institutional policies governing project allocation*
- *Timely participation from students and faculty*

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a responsive, web-based graphical user interface (GUI) accessible through standard web browsers. The user interface shall be role-specific and dynamically adapt based on the authenticated user's role (Student, Faculty, Department Coordinator, System Administrator).

General UI Characteristics

- *Clean, intuitive, and consistent layout across all pages*
- *Role-based dashboards after login*
- *Standard navigation menu, logout option, and help section on all screens*
- *Form validation with clear error messages*
- *Confirmation dialogs for critical actions (apply, accept, finalize)*

Student Interface

- *Profile creation and update page*
- *Team creation and team management interface (max 3 members)*
- *Team Leader dashboard showing:*
    - *Applied faculty/projects*
    - *Faculty acceptances*
    - *Option to confirm or wait*
- *Faculty listing and project browsing page*
- *Notifications panel for application status and responses*

Faculty Interface

- *Faculty profile management page*
- *Project creation and management interface*
- *Dashboard displaying:*
    - *List of teams who applied*
    - *Team details (members, resumes, CGPA, department)*
    - *Accept/Reject options*
- *Slot availability display*

Department Coordinator / Admin Interface

- *System configuration dashboard*
- *Timeline and slot management screens*
- *Allocation monitoring interface*
- *Override and reassignment controls*
- *Report generation interface*

System Administrator Interface

- *User and role management screens*
- *System logs and audit trail viewer*
- *Security and access control management*
- *Backup and maintenance controls*

*The UI shall be developed using **React** and must be responsive to support different screen sizes.*

3.2 Hardware Interfaces

The system does not require any specialized hardware interfaces. It shall operate on standard computing devices.

Supported Hardware

- *Client devices:*
  - *Desktop computers*
  - *Laptops*
  - *Tablets*
- *Server hardware:*
  - *Standard application server with sufficient CPU, RAM, and storage*

Hardware Interaction

- *Input devices: Keyboard, Mouse, Touchscreen*
- *Output devices: Monitor, Printer (for reports)*

All hardware interaction will be handled through the operating system and web browser.

3.3 Software Interfaces

The system will interface with the following software components:

Operating Systems

- *Windows 10 and above*
- *Linux (Ubuntu, CentOS, etc.)*
- *macOS (for client access)*

Web Browsers

- *Google Chrome (latest version)*

- *Mozilla Firefox*
- *Microsoft Edge*
- *Safari*

Backend Framework

- ***Java Spring Boot** – REST API services, business logic, security*

Frontend Framework

- ***React.js** – user interface and client-side logic*

Database

- *MariaDB (Relational Database)*

Authentication System

- *Institutional login system or custom authentication module*
- *Role-based access control (RBAC)*

File Handling

- *Resume/CV upload and storage system*

Notification Services

- *Email server (SMTP) for alerts and confirmations (if applicable)*

*All interactions between frontend and backend shall use **Resultful APIs** with JSON data format*

3.4 Communications Interfaces

The system shall use standard web communication protocols.

Protocols

- ***HTTP / HTTPS** for client-server communication*
- ***TCP/IP** for network transmission*

Data Format

- *JSON for API requests and responses*
- *Multipart/form-data for file uploads (resumes)*

Security

- *HTTPS for secure communication*
- *Token-based authentication (e.g., JWT)*
- *Encrypted password storage*
- *Session management and timeout handling*

Notifications

- *In-app notifications for:*
    - *Application submission*
    - *Faculty acceptance/rejection*
    - *Final confirmation*
- *Optional email notifications for important events*

Network Requirements

- *Requires stable internet or institutional network connectivity*
- *Should handle multiple concurrent users*

4. System Features

This section describes the major functional features of the UG/PG Project Allocation System.

Each feature is specified with its description, priority, stimulus/response sequence, and detailed functional requirements.

4.1 User Authentication and Role Recognition

Description and Priority

Description:

This feature authenticates users using institutional credentials and identifies their role within the system. Based on the recognized role, appropriate permissions and dashboards are assigned.

Priority: *High*

Stimulus / Response Sequence

1. *User enters institutional credentials.*

2. *System validates credentials.*
3. *System identifies the associated role.*
4. *System grants access and loads the role-specific dashboard.*

Functional Requirements

FR-4.1-1: The system shall authenticate users using institutional credentials.

FR-4.1-2: The system shall identify the user role upon successful authentication.

FR-4.1-3: The system shall deny access for invalid credentials with an error message.

4.2 Role-Based Dashboard Access

Description and Priority

Description:

This feature ensures that users can access only those system functions permitted by their role.

Priority: *High*

Stimulus / Response Sequence

1. *User logs in successfully.*
2. *System checks role permissions.*
3. *System loads the appropriate dashboard.*

Functional Requirements

FR-4.2-1: The system shall provide separate dashboards for students, faculty, department coordinators, and system administrators.

FR-4.2-2: The system shall restrict access to unauthorized features.

4.3 Project Issuance by Faculty

Description and Priority

Description:

This feature allows faculty members to create and publish project offerings for student selection.

Priority: *High*

Stimulus / Response Sequence

1. *Faculty creates a project.*
2. *Faculty specifies domain and slot capacity.*
3. *Faculty publishes the project.*
4. *System makes the project visible to students.*

Functional Requirements

FR-4.3-1: The system shall allow faculty to create project entries.

FR-4.3-2: The system shall associate each project with a faculty member and domain.

FR-4.3-3: The system shall allow faculty to publish or withdraw projects.

4.4 Project Listing and Discovery (Student View)

Description and Priority

Description:

This feature enables students to view all currently available faculty-issued projects.

Priority: *High*

Stimulus / Response Sequence

1. *Student opens the dashboard.*

2. *System displays the list of published projects.*

Functional Requirements

FR-4.4-1: The system shall display only published projects to students.

FR-4.4-2: The system shall hide unpublished or withdrawn projects.

4.5 Project Filtering and Search

Description and Priority

Description:

This feature allows students to filter and search projects efficiently.

Priority: *Medium*

Stimulus / Response Sequence

1. *Student selects filter criteria.*
2. *System updates the project list accordingly.*

Functional Requirements

FR-4.5-1: The system shall support filtering by domain.

FR-4.5-2: The system shall support filtering by faculty name and department.

FR-4.5-3: The system shall support filtering by slot availability.

4.6 Team Formation and Management

Description and Priority

Description:

This feature allows students to form teams for project applications.

Priority: *High*

Stimulus / Response Sequence

1. *Student creates a team.*
2. *Student adds members.*
3. *System validates team constraints.*

Functional Requirements

FR-4.6-1: The system shall allow students to create teams.

FR-4.6-2: The system shall restrict students to one active team.

FR-4.6-3: The system shall validate team size.

4.7 Application Submission

Description and Priority

Description:

This feature allows teams to apply for faculty projects.

Priority: *High*

Stimulus / Response Sequence

1. *Team lead selects a project.*
2. *Team lead submits application.*

3. *System validates the application.*

Functional Requirements

FR-4.7-1: The system shall allow only team leads to submit applications.

FR-4.7-2: The system shall prevent applications after deadlines.

4.8 Application Review and Notes (Faculty)

Description and Priority

Description:

This feature allows faculty to review applications and record internal notes.

Priority: *Medium*

Stimulus / Response Sequence

1. *Faculty opens an application.*
2. *Faculty writes notes.*
3. *System saves the notes.*

Functional Requirements

FR-4.8-1: The system shall allow faculty to view submitted applications.

FR-4.8-2: The system shall allow faculty to add notes to applications.

4.9 Read / Unread Application Tracking

Description and Priority

Description:

This feature tracks whether applications have been viewed by faculty.

Priority: *Medium*

Stimulus / Response Sequence

1. *Faculty opens an application.*
2. *System marks it as read.*

Functional Requirements

FR-4.9-1: The system shall mark applications as unread by default.

FR-4.9-2: The system shall update status when viewed.

4.10 Acceptance and Rejection Workflow

Description and Priority

Description:

This feature allows faculty to accept or reject teams.

Priority: *High*

Stimulus / Response Sequence

1. *Faculty selects decision.*
2. *System updates application status.*
3. *Team is notified.*

Functional Requirements

FR-4.10-1: The system shall allow faculty to accept applications.

FR-4.10-2: The system shall allow faculty to reject applications.

4.11 Dynamic Slot Management

Description and Priority

Description:

This feature maintains correct slot availability for projects.

Priority: *High*

Stimulus / Response Sequence

1. *Application is confirmed.*
2. *System reduces slot count.*
3. *Project is marked full if limit reached.*

Functional Requirements

FR-4.11-1: The system shall decrement slots upon confirmation.

FR-4.11-2: The system shall prevent confirm beyond capacity.

4.12 Team Status Management

Description and Priority

Description:

This feature tracks the status of teams throughout the workflow.

Priority: *High*

Functional Requirements

FR-4.12-1: The system shall maintain team states such as Applied, Accepted, Confirmed, and Rejected.

4.13 Student Status Tracking

Description and Priority

Description:

This feature provides students with transparency on application outcomes.

Priority: *High*

Functional Requirements

FR-4.13-1: The system shall display application and allocation status to students.

4.14 Faculty Status Tracking

Description and Priority

Description:

This feature allows faculty to track supervision load.

Priority: *Medium*

Functional Requirements

FR-4.14-1: The system shall display filled and remaining slots to faculty.

4.15 User Profile Viewing

Description and Priority

Description:

This feature allows users to view basic profile information.

Priority: *Low*

Functional Requirements

FR-4.15-1: The system shall display name, role, department, and email.

4.16 Email Redirection and Communication

Description and Priority

Description:

This feature enables direct email communication through redirection.

Priority: *Low*

Functional Requirements

FR-4.16-1: The system shall provide an email compose link.

4.17 Department Coordinator Monitoring Dashboard

Description and Priority

Description:

This feature provides coordinators with a real-time overview of allocations.

Priority: *High*

Functional Requirements

FR-4.17-1: The system shall display all teams and faculty allocations dynamically.

4.18 Faculty Capacity Control

Description and Priority

Description:

This feature allows coordinators to define faculty supervision limits.

Priority: *High*

Functional Requirements

FR-4.18-1: The system shall enforce maximum project limit per faculty.

4.19 Deadline Management

Description and Priority

Description:

This feature controls time-bound activities in the allocation process.

Priority: *High*

Functional Requirements

FR-4.19-1: The system shall restrict submissions after deadlines.

4.20 Audit and Activity Tracking

Description and Priority

Description:

This feature logs critical system actions.

Priority: *Low*

Functional Requirements

FR-4.20-1: The system shall log allocation and configuration actions.

4.21 Error Handling and Invalid Scenarios

Description and Priority

Description:

This feature defines system behaviour for invalid operations.

Priority: *Medium*

Functional Requirements

FR-4.21-1: The system shall display meaningful error messages for invalid actions.

4.22 Assumptions and Constraints

Description

- *Institutional user data is pre-existing.*
- *Email communication is handled externally.*
- *Each system instance serves a single department.*

4.23 Project Duration and Multi-Semester Allocation Handling

Description and Priority

Description:
This feature allows faculty members to specify the duration of a project at the time of creation. The selected duration determines whether students allocated to the project are required to participate in the project allocation process in subsequent semesters. Students working on projects with a duration of more than one semester are automatically exempted from applying again until the project duration is completed.

The selected duration is visible to students and the department coordinator.

Priority: *High*

Stimulus / Response Sequence

1. *Faculty creates a project and selects its duration.*
2. *Faculty publishes the project.*
3. *Team is allocated to the project.*
4. *System records the project duration for the allocated team.*
5. *At the start of a new semester:*
    a. *System checks ongoing project duration for each student.*
    b. *Students with remaining project duration are exempted from the allocation process.*
6. *System restricts exempted students from submitting new applications until the project duration ends.*

Functional Requirements

FR-4.23-1: The system shall allow faculty to select a project duration while creating a project.
FR-4.23-2: The system shall support predefined project durations.
FR-4.23-3: The system shall associate the selected project duration with the allocated team.
FR-4.23-4: The system shall track remaining project duration across semesters.
FR-4.23-5: The system shall exempt students working on multi-semester projects from applying in subsequent semesters until the project duration is completed.
FR-4.23-6: The system shall prevent exempted students from submitting new project applications.
FR-4.23-7: The system shall display project duration and exemption status to the department coordinator.
FR-4.23-8: The system shall allow students to view the duration and remaining semesters of their allocated project.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall meet the following performance requirements to ensure responsive, reliable, and efficient operation under normal and peak usage conditions. These requirements are intended to support smooth interaction between students and faculty during project allocation workflows, including team formation and the three-way approval process.

1. ***Response Time***

    - *The system shall respond to all user-initiated requests (such as login, viewing projects, submitting project applications, and accepting/rejecting requests) within **2 seconds** under normal operating conditions.*

    - *In peak usage scenarios (e.g., during project allocation deadlines), the system shall respond within **5 seconds** for at least **95% of requests**.*

2. ***Concurrent User Handling***

    - *The system shall support a minimum of **500 concurrent users** without significant degradation in performance.*

    - *Concurrent access shall include simultaneous student logins, team formation activities, and faculty review of project applications.*

3. ***Transaction Processing***

    - *The system shall process critical transactions such as:*

        - *Team creation and validation (maximum team size of 3),*

        - *Project application submission by team leaders,*

        - *Faculty acceptance or rejection of applications,*

        - *Student confirmation of faculty acceptance*

    *in a **consistent and atomic manner**, ensuring no partial or inconsistent states occur during the three-way handshake process.*

4. ***Data Retrieval Performance***

- *The system shall retrieve and display lists of projects, faculty details, and application statuses within **2 seconds** for standard query operations.*

- *Filtering and sorting operations on project or faculty listings shall complete within **3 seconds**.*

5. ***Backend Processing Efficiency***

    - *The backend services implemented using Spring Boot shall handle business logic and database operations efficiently, ensuring minimal latency between request reception and response generation.*

    - *Asynchronous processing shall be used where applicable (e.g., notifications or status updates, automated emails) with Redis-Cache to avoid blocking user interactions.*

6. ***Scalability Considerations***

    - *The system shall be designed to allow horizontal scaling of backend services to accommodate increased user load in future academic years without major architectural changes.*

These performance requirements ensure that the system provides a smooth and reliable user experience for both students and faculty, particularly during critical academic timelines, while supporting future growth and maintainability.

5.2 Safety Requirements

The system shall incorporate safeguards to prevent data loss, incorrect project allocation, unauthorized actions, and misuse of system privileges that could adversely affect students, faculty, or institutional academic processes. Although the system does not involve physical safety risks, ensuring correctness, integrity, and controlled administrative intervention is critical to avoid academic and operational harm.

1. ***Data Integrity and Loss Prevention***

    - *The system shall protect all critical data, including user profiles, team details, project applications, and allocation decisions, from accidental loss or corruption.*

    - *All allocation-related transactions shall be executed atomically to maintain consistency throughout the three-way handshake process.*

2. ***Prevention of Invalid Operations***

    - *The system shall prevent students from forming teams exceeding the maximum allowed team size of **three members**.*

    - *The system shall restrict project application submission to **team leaders only**.*

    - *The system shall prevent duplicate or conflicting project allocations for the same team.*

3. ***Controlled Allocation Workflow***

    - *The system shall enforce the three-way handshake mechanism such that project allocation is finalized only after:*

        - *Submission of an application by the student team leader,*

        - *Acceptance by the concerned faculty member,*

        - *Confirmation by the student team leader.*

    - *The system shall prevent automatic or forced allocation without explicit consent from both academic parties.*

4. ***Failure Handling and Recovery***

    - *In the event of system failure during critical operations, the system shall rollback incomplete transactions to prevent inconsistent allocation states.*

    - *The system shall provide clear error messages and safe recovery options to users following failed operations.*

5. ***Administrative Safeguards***

    - *The system shall allow the **Department Coordinator** to intervene in exceptional or conflict scenarios to prevent unsafe or incorrect project allocations.*

    - *Administrative interventions shall be controlled, logged, and limited to authorized circumstances.*

6. ***Access Control Safety***

- *The system shall restrict access to sensitive operations and data based on assigned user roles (Student, Faculty, Department Coordinator, System Administrator).*

- *Unauthorized users shall be prevented from viewing, modifying, or interfering with project allocation data.*

7. ***Audit and Traceability***

- *The system shall maintain audit logs for critical actions, including application submission, faculty decisions, student confirmations, and administrative interventions.*

- *Audit records shall be protected against unauthorized modification and deletion.*

8. ***Policy Compliance***

- *The system shall operate in accordance with institutional academic policies governing UG/PG project allocation.*

- *Safety-related system behaviour shall comply with applicable institutional data protection and governance guidelines.*

These safety requirements ensure that the system minimizes risks related to incorrect project allocation, data loss, and unauthorized actions, thereby safeguarding the academic interests of all stakeholders.

5.3 Security Requirements

The system shall implement security controls to protect user identities, academic data, and system resources from unauthorized access, misuse, or disclosure. These requirements ensure confidentiality, integrity, and accountability across all user roles within the UG/PG Project Allocation System.

1. ***User Authentication***

- *The system shall require all users to authenticate using valid institutional credentials before accessing any system functionality.*

- *Each authenticated user shall be uniquely identified and associated with exactly one role: Student, Faculty, Department Coordinator, or System Administrator.*

- *The system shall prevent unauthenticated or anonymous access to protected system resources.*

2. ***Role-Based Access Control (RBAC)***

    - *The system shall enforce role-based access control to restrict system operations based on assigned roles.*
    - *Students shall only be permitted to access and manage their own profiles, team details, and project applications.*
    - *Faculty members shall only be permitted to access project applications submitted to projects they supervise.*
    - *Department Coordinators shall be permitted to monitor the allocation process, manage allocation timelines, and intervene in exceptional cases.*
    - *System Administrators shall be permitted to manage system infrastructure, user accounts, and security configurations, without access to academic decision-making functions.*

3. ***Data Confidentiality and Privacy***

    - *The system shall protect sensitive data, including user credentials, academic records, and project allocation details, from unauthorized disclosure.*
    - *Authentication credentials shall not be stored or transmitted in plain text.*
    - *Access to personal and academic data shall be limited to authorized users based on role and data ownership.*

4. ***Secure Communication***

    - *The system shall ensure that all communication between the client interface (React frontend) and backend services (Spring Boot) occurs over secure communication channels.*
    - *The system shall prevent interception, tampering, or replay of data during transmission.*

5. ***Authorization and Action Validation***

    - *The system shall validate every security-sensitive operation to ensure that the requesting user has sufficient privileges.*

- *The system shall prevent users from:*
  - *Submitting applications on behalf of other teams,*
  - *Accessing or modifying allocations outside their authorized scope,*
  - *Altering finalized project allocation records without administrative authorization.*

6. ***Audit Logging and Accountability***
   - *The system shall maintain secure audit logs of security-relevant events, including login attempts, application submissions, acceptances, confirmations, and administrative actions.*
   - *Audit logs shall be accessible only to authorized administrative roles and protected against unauthorized modification or deletion.*
7. ***Administrative Security Controls***
   - *The system shall restrict administrative privileges to authorized Department Coordinators and System Administrators only.*
   - *All administrative security actions shall be logged to ensure accountability and traceability.*
8. ***Compliance with Policies and Best Practices***
   - *The system shall comply with institutional information security policies and applicable data protection guidelines.*
   - *The system shall follow established best practices for secure web application development and deployment.*

These security requirements ensure that the system safeguards academic data, enforces controlled access, and maintains user privacy throughout the project allocation process.

5.4 Software Quality Attributes

The system shall exhibit the following software quality attributes to ensure reliability, usability, and maintainability for both end users and developers. These attributes are prioritized to support academic workflows, scalability, and long-term system evolution.

1. ***Usability***

    - *The system shall provide an intuitive and user-friendly interface that enables students and faculty to perform core tasks with minimal training.*
    - *Common user operations such as login, team formation, project application, and acceptance shall be completable within **three user interactions or fewer**.*
    - *The system shall prioritize **ease of use over ease of learning**, ensuring straightforward navigation and clear feedback messages.*

2. ***Reliability***

    - *The system shall maintain correct functionality during normal and peak usage periods, particularly during project allocation deadlines.*
    - *The system shall achieve an operational availability of at least **99%** during the academic allocation period.*
    - *The system shall ensure consistent system states during the three-way handshake process, avoiding partial or conflicting allocations.*

3. ***Availability***

    - *The system shall be accessible to users at all times except during scheduled maintenance.*
    - *Scheduled maintenance downtime shall be communicated in advance and shall not exceed **2 hours per occurrence**.*

4. ***Maintainability***

    - *The system shall be designed using modular architecture, separating frontend (React) and backend (Spring Boot) components to simplify maintenance.*
    - *Changes to business rules (e.g., team size limits or allocation workflows) shall be implementable without requiring major system redesign.*

5. ***Scalability***

    - *The system shall support an increase in the number of users, projects, and applications without significant performance degradation.*

- *The system shall allow backend services to be scaled horizontally to handle future academic batches.*

6. ***Interoperability***

    - *The system shall interact with standard web browsers and institutional systems using widely accepted web standards and protocols.*

    - *The system shall support integration with existing institutional authentication or academic management systems, if required in the future.*

7. ***Security and Privacy (Quality Perspective)***

    - *The system shall ensure confidentiality, integrity, and controlled access to academic and personal data.*

    - *Security measures shall not significantly degrade system usability or performance.*

8. ***Testability***

    - *The system shall be designed to support unit testing, integration testing, and system testing.*

    - *Critical workflows such as project application submission and allocation confirmation shall be verifiable through automated test cases.*

These quality attributes ensure that the system is dependable, easy to use, and adaptable to future requirements while meeting both user and developer expectations.

5.5 Business Rules

The system shall adhere to the following business rules, which define the operating principles governing user roles, responsibilities, and interactions within the UG/PG Project Allocation System. These rules ensure consistent enforcement of institutional policies and proper execution of the project allocation process.

1. ***User Role Classification***

    - *Every system user shall be assigned exactly one primary role: Student (UG/PG), Faculty/Professor, Department Coordinator/Admin, or System Administrator.*

- *User permissions and accessible functionalities shall be determined solely based on the assigned role.*

2. ***Student Role Rules***

- *Only students enrolled in undergraduate or postgraduate programs shall be permitted to register as Student users.*
- *Students may form teams with a maximum size of **three members**.*
- *Each team shall have exactly one designated **team leader**.*
- *Only the team leader shall be permitted to apply for projects on behalf of the team.*
- *Team members other than the leader shall not be permitted to submit or modify project applications.*

3. ***Faculty Role Rules***

- *Faculty members shall be permitted to create and manage their professional profiles, including areas of expertise and supervision details.*
- *Faculty members shall only be permitted to view and review project applications submitted to projects they offer.*
- *Faculty decisions to accept or reject applications shall not result in final allocation until confirmed by the student team leader.*

4. ***Three-Way Allocation Handshake Rule***

- *A project allocation shall be considered complete only after:*

    1. *The student team leader submits an application,*
    2. *The faculty member accepts the application,*
    3. *The student team leader confirms the faculty acceptance.*

- *Until all three steps are completed, the allocation shall remain in a pending state.*

5. ***Department Coordinator / Admin Rules***

- *The Department Coordinator shall oversee the overall project allocation process across departments.*
- *The Department Coordinator shall have the authority to monitor allocation progress, manage system-level settings related to allocation timelines, and intervene in case of conflicts or exceptional situations.*
- *The Department Coordinator shall not directly modify individual student or faculty profiles unless required for administrative resolution.*

6. ***System Administrator Rules***

- *The System Administrator shall be responsible for managing system infrastructure, user accounts, and security configurations.*
- *The System Administrator shall not participate in academic decision-making related to project allocation.*

- *Administrative actions performed by the System Administrator shall be logged for accountability.*

7. ***Conflict and Exception Handling***

- *In case of conflicting allocations or unresolved application states, administrative intervention by the Department Coordinator shall take precedence over automated system decisions.*

- *Final authority for resolving exceptional cases shall rest with the Department Coordinator, in accordance with institutional policies.*

These business rules define the governance structure of the system and guide the implementation of functional requirements while ensuring transparency, fairness, and policy compliance.

6. Other Requirements

This section specifies additional requirements that do not fall directly under functional or non-functional categories but are essential for the successful deployment, operation, and future evolution of the UG/PG Project Allocation System.

6.1 Database Requirements

- *The system shall use a centralized relational database to store user profiles, team information, project details, applications, and allocation records.*

- *The database shall enforce referential integrity among entities such as students, teams, faculty, and projects.*

- *The database shall support transactional consistency to maintain correct system states during the project allocation workflow.*

- *Regular database backups shall be maintained to prevent data loss in case of system failure.*

6.2 Deployment and Platform Requirements

- *The frontend shall be implemented using **React** and accessed through standard web browsers.*

- *The backend shall be implemented using **Java with Spring Boot**.*

- *The system shall be deployable on institutional servers or cloud-based infrastructure.*
- *The system shall support deployment on commonly used operating systems without requiring platform-specific modifications.*

6.3 Browser and Device Compatibility

- *The system shall support all modern web browsers commonly used in academic environments.*
- *The user interface shall be responsive and usable on desktops, laptops, and tablets.*
- *Mobile device support shall be provided where feasible, without compromising usability.*

6.4 Legal and Policy Requirements

- *The system shall comply with institutional academic policies governing UG/PG project allocation and supervision.*
- *The system shall adhere to applicable data protection and privacy regulations enforced by the institution.*
- *User data shall be used strictly for academic and administrative purposes.*

6.5 Logging and Monitoring Requirements

- *The system shall log critical system and user activities for monitoring, auditing, and troubleshooting purposes.*
- *System logs shall be protected from unauthorized access and modification.*
- *Administrative users shall be able to review logs relevant to system operations and security events.*

6.6 Reusability and Extensibility

- *The system shall be designed with modular components to facilitate reuse in other academic management systems.*
- *The system shall support future extensions, such as additional allocation rules or integration with institutional portals, with minimal architectural changes.*

6.7 Data Retention and Archival

- *The system shall retain project allocation records for a configurable academic period as defined by institutional policy.*
- *Archived data shall remain accessible to authorized administrative users for reporting and verification purposes.*

6.8 Assumptions and Dependencies

- *The system assumes availability of reliable internet connectivity for all users.*
- *The system depends on the accuracy of institutional data provided for user registration and role assignment.*
- *Changes in institutional policies may require corresponding system updates.*

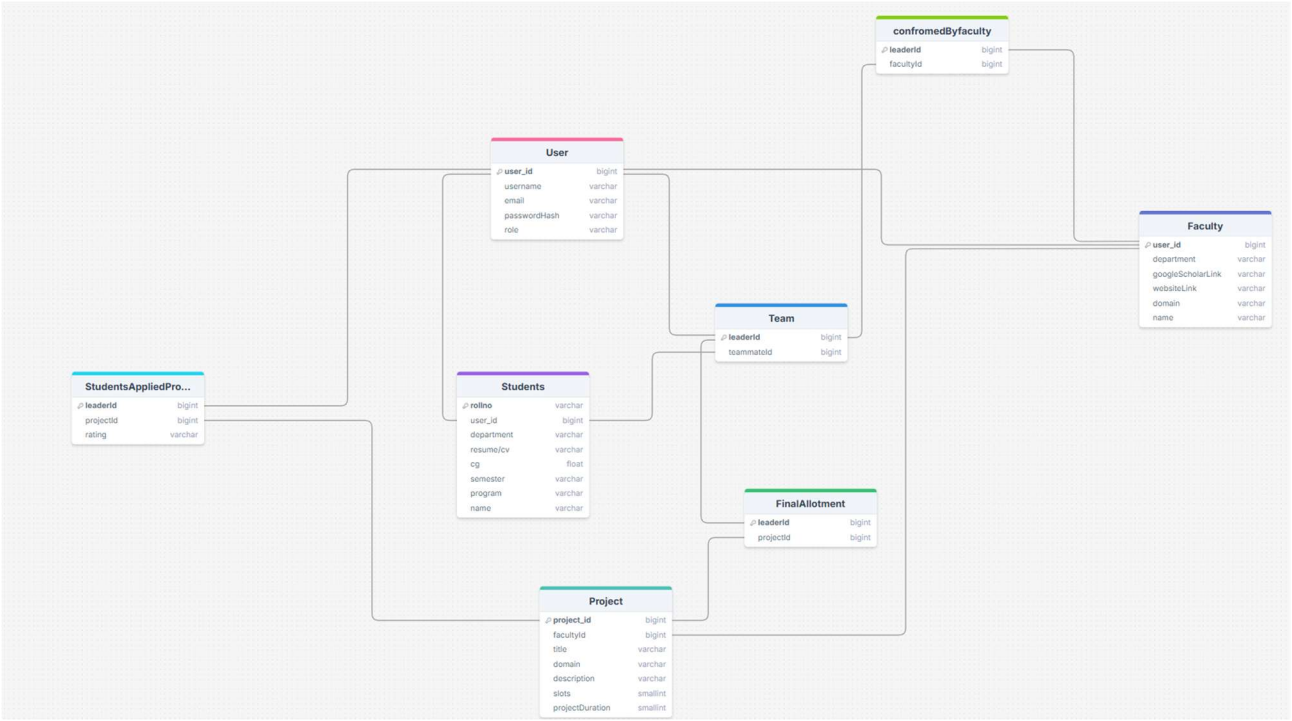
Appendix A: Glossary

This glossary defines the terms, acronyms, and abbreviations used in this Software Requirements Specification (SRS) to ensure a clear and consistent understanding of the UG/PG Project Allocation System.

Term / Acronym	***Definition***
UG	*Undergraduate program offered by the institution.*
PG	*Postgraduate program offered by the institution.*
Project Allocation System	*A web-based software system used to allocate academic projects to student teams under faculty supervision.*
Student	*A registered undergraduate or postgraduate user of the system who participates in project allocation.*
Faculty / Professor	*An academic staff member who offers and supervises student projects.*
Department Coordinator / Admin	*An authorized academic administrator responsible for overseeing and managing the project allocation process at the department level.*
System Administrator	*A technical user responsible for managing system infrastructure, user accounts, security configurations, and maintenance.*
Team	*A group of students collaborating on a project, consisting of a maximum of three members.*
Team Leader	*A designated student within a team who is authorized to submit project applications and confirm allocations on behalf of the team.*
Three-Way Handshake	*The controlled allocation workflow consisting of three steps: team application submission, faculty acceptance, and team confirmation.*

Term / Acronym	***Definition***
Project	*An academic task or research topic offered by a faculty member for student supervision.*
Application	*A formal request submitted by a team leader to a faculty member for project supervision.*
Allocation	*The final assignment of a project to a student team after completion of the three-way handshake process.*
RBAC (Role-Based Access Control)	*A security mechanism that restricts system access and operations based on user roles.*
Authentication	*The process of verifying a user's identity before granting access to the system.*
Authorization	*The process of granting or denying access to specific system functions based on user roles.*
React	*A JavaScript-based frontend framework used to build the user interface of the system.*
Spring Boot	*A Java-based backend framework used to implement business logic and RESTful services.*
REST API	*Representational State Transfer Application Programming Interface used for client–server communication.*
MariaDB	*A relational database management system used for persistent data storage.*
Audit Log	*A record of system and user activities maintained for security, monitoring, and accountability purposes.*
Institutional Credentials	*Official login credentials provided by the academic institution to users.*
Slot	*The supervision capacity allocated by a faculty member for student projects.*
Deadline	*A predefined time limit after which certain system actions (e.g., applications) are restricted.*

Appendix B: Analysis Models



Appendix C: To Be Determined List

The following items are identified as To Be Determined (TBD) within this Software Requirements Specification. These items require further clarification, institutional decision-making, or implementation-time configuration and will be tracked to closure during later phases of system design and deployment.

1. ***TBD-1: Institutional Authentication Mechanism***
   The exact authentication mechanism (integration with an existing institutional Single Sign-On system or use of a custom authentication module) is yet to be finalized.

2. ***TBD-2: Maximum Faculty Supervision Capacity Defaults***
   The default maximum number of projects or student teams that a faculty member may supervise has not been fixed and will be determined by departmental policy.

3. ***TBD-3: Project Duration Options***
   The predefined list of supported project durations (e.g., 1 semester, 2 semesters, 1 academic year) requires confirmation from academic authorities.

4. ***TBD-4: Allocation Timelines and Deadlines***
   Exact dates and duration for application submission, faculty acceptance, and team confirmation phases will be configured per academic cycle.

5. ***TBD-5: Email Notification Enablement***
   Whether email notifications are mandatory, optional, or disabled by default depends on availability of institutional SMTP services and policy decisions.

6. ***TBD-6: Data Retention Period***
   The duration for which project allocation records, audit logs, and historical data must be retained before archival or deletion is subject to institutional policy.

7. ***TBD-7: Administrative Override Rules***
   Specific conditions and limitations under which Department Coordinators may override allocations require formal policy definition.

8. ***TBD-8: Backup Frequency and Recovery Policy***
   The frequency of database backups and acceptable recovery time objectives (RTO) will be finalized during deployment planning.

9. ***TBD-9: Mobile Device Support Scope***
   The extent of functional support on mobile devices (read-only vs full functionality) is yet to be determined.

10. ***TBD-10: Integration with Other Institutional Systems***
    Future integration with existing academic management systems (e.g., student information systems or learning management systems) is currently undefined.